# Invariant points and lines

## Key Points

For a matrix transform $M$ invariant points remain at the same point.

That is $\binom{x}{y}$ is an invariant point if

$$M\binom{x}{y} = \binom{x}{y}$$

An invariant line is such that all points on the line are transformed to other points on the line. To find the line use $y = mx + c$ and $y' = mx' + c$ so

$$M\binom{x}{mx+c} = \binom{x'}{mx'+c}$$

Note. An invariant line is not necessarily a line of invariant points

## Basic Skills

**Q1.** For the matrix transform $M = \begin{pmatrix} 3 & 4 \\ 5 & -2 \end{pmatrix}$

**a)** Find out what happens to the point $\binom{2}{-1}$ under the transform $M$

**b)** Find out what happens to the point $\binom{x}{y}$ under the transform $M$

**c)** Find out what point becomes $\binom{1}{1}$ under $M$

**Q2.** For the matrix transform $M = \begin{pmatrix} -3 & 2 \\ 0 & 1 \end{pmatrix}$ write down an equation for the new coordinates $\binom{x'}{y'}$ in terms of the old coordinates $x, y$

**Q3.** Explain why the origin is always and invariant point

## Competency Skills

**Q4.** For the matrix $M = \begin{pmatrix} -3 & 4 \\ 3 & -2 \end{pmatrix}$ verify that the point (1,1) is an invariant point under the transform represented by $M$

**Q5.** For the matrix $M = \begin{pmatrix} 3 & -1 \\ 2 & 0 \end{pmatrix}$ show that all points on the line $y = 2x$ are invariant points and decide if the line is invariant

**Q6.** For $M = \begin{pmatrix} 4 & 2 \\ 6 & 5 \end{pmatrix}$ show that $y = 2x - 3$ is an invariant line

**Q7.** Do a line of invariant points form an invariant line and vice versa

**Q8.** Find the invariant lines of the transform represented by $M = \begin{pmatrix} 4 & -1 \\ -2 & 1 \end{pmatrix}$

## Advanced Skills

**Q9.** For the transform given by matrix $M = \begin{pmatrix} 5 & 2 \\ 2 & 5 \end{pmatrix}$

**a)** Show that the only invariant point is the origin

**b)** Find the invariant lines of the transform

**Q10.** Determine if there is any invariance for a rotation of 45° anti-clockwise around the origin

**Q11.** By using the general transforms matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ determine the conditions for

**a)** No invariant lines

**b)** Infinite invariant lines

## Extension

**Q13.** Using the matrix $\begin{pmatrix} \frac{1-m^2}{1+m^2} & \frac{2m}{1+m^2} \\ \frac{2m}{1+m^2} & \frac{m^2-1}{1+m^2} \end{pmatrix}$ which represents a reflection in $y = mx$ show that the line of reflection is invariant